Djkshtra Algorithm and Distance Vector Routing

Why Routing Algorithms?

- Routing decides **best path for data packets**
- Goal:
  - Minimum cost
  - Fast delivery
  - Loop-free routing
- Two main approaches:
  - **Distance Vector Routing**
  - **Link State Routing (Dijkstra)**

Distance Vector Routing(DVR)

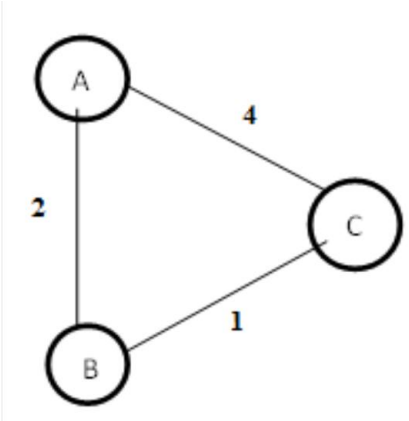
- Each router maintains a **distance vector table**
- Table contains:
  - Destination
  - Distance (cost)
  - Next hop
- Routers **share tables only with neighbors**
- Periodic updates
- *Based on Bellman–Ford Algorithm*

DVR(Problems)

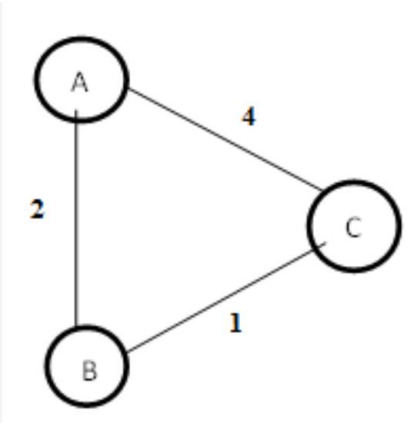
- Slow convergence
- Routing loops
- Count-to-Infinity problem
- Limited network knowledge
- Example protocol: **RIP**

Working (DVR)

Step 1 − In this DVR network, each router shares its routing table with every neighbor.
For example, A will share its routing table with neighbors B and C and neighbors B and C will share their routing table with A.



Form A	A	B	C
A	0	2	3
B			
C			



Form A	A	B	C
A	0	2	3
B			
C			

Form B	A	B	C
A			
B	2	0	1
C			

Form C	A	B	C
A			
B			
C	3	1	0

Step 2 − If the path via a neighbor has a lower cost, then the router updates its local table to forward packets to the neighbor. In this table, the router updates the lower cost for A and C by updating the new weight from 4 to 3 in router A and from 4 to 3 in router C.

Step 3 − The final updated routing table with lower cost distance vector routing protocol for all routers A, B, and C is given below −

Form A	A	B	C
A	0	2	3
B	2	0	1
C	3	1	0

Form B	A	B	C
A	0	2	3
B	2	0	1
C	3	1	0

Form C	A	B	C
A	0	2	3
B	2	0	1
C	3	1	0

Count-to-Infinity Problem

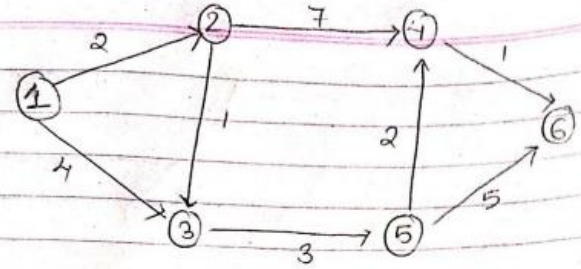
- Occurs when a node become unreachable.
- Routers keep increasing hop count.
- Happens due to wrong neighbor information
- RIP defines infinity as 16 hops.

Dijkshtra Algorithm

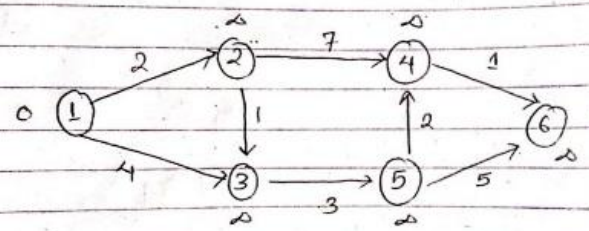
- Used in Linked State Routing.
- Finds **shortest path from source to all nodes.**
- Each router knows **entire network topology.**
- Uses **greedy approach.**

Working(Dijkshtra Algorithm)

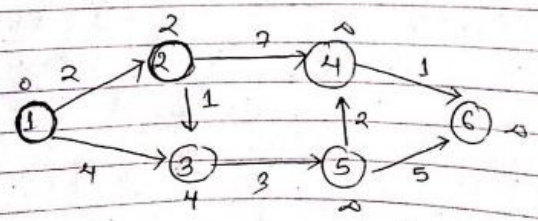
- Assign distance = 0 to source, ∞ to others
- Select unvisited node with **minimum distance**
- Mark it visited
- Relax its neighbors
- Repeat until all nodes are visited
- *Once a node is visited, its distance is **final***



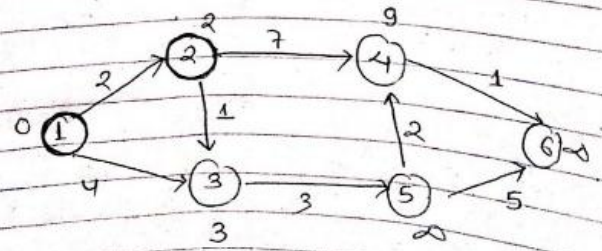
Step (1) let Source vertex is 1



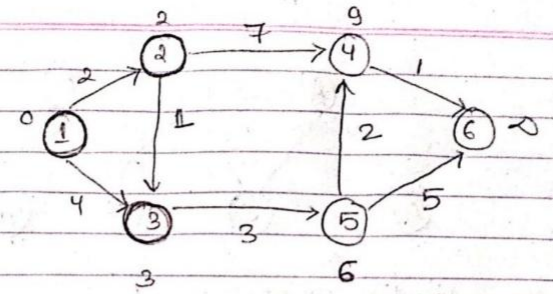
Step (2)



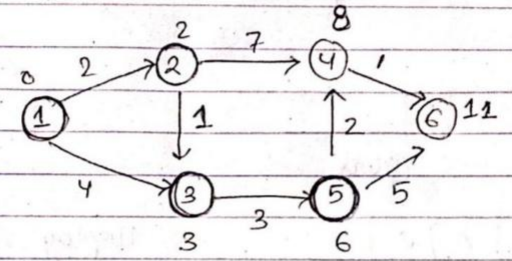
Step (3)



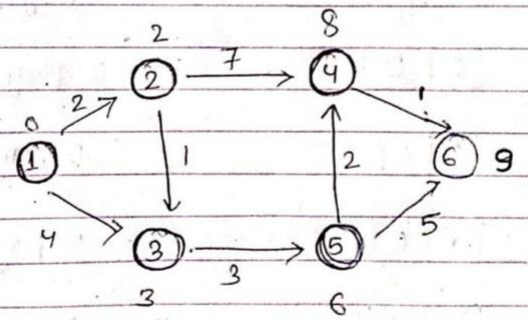
Step(4)



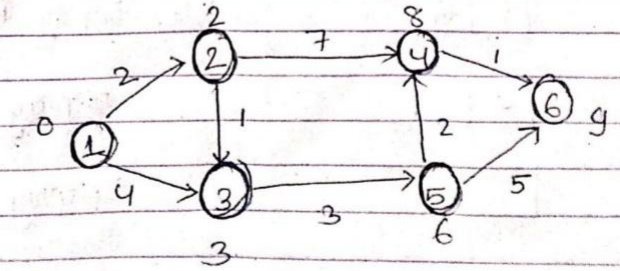
Step (5)



Step (6)



Step (7)



Dijkshtra Algorithm(Features)

- Fast convergence
- No routing loops
- Cannot handle negative edge weights
- Higher memory and CPU Usage

Comparison

Feature	Distance Vector	Dijkstra
Algorithm	Bellman–Ford	Dijkstra
Network knowledge	Neighbors only	Full topology
Convergence	Slow	Fast
Count-to-infinity	Yes	No
Example	RIP	OSPF

Advantages and Disadvantages

- **Distance Vector**
  - Simple
  - Low overhead
  - Slow & unstable
- **Dijkstra**
  - Accurate & fast
  - Scalable
  - Complex & costly

Thank you